

ARP QUESTION

- 1) **fork() and exec() are similar because (choose the answer that you consider most complete and significant)**
 - they preserve many information owned by the parent and given to the child after fork / exec
 - they preserve open file descriptors shared by the two processes after fork and father / son after exec
- 2) **The most important difference between unnamed and named pipes is: (choose the most complete and significant)**
 - named pipes can be used by several processes simultaneously
- 3) **Choose RIGHT and significant sentences about the switcher (select multiple answers)**
 - is a routine called by an interrupt or a signal
 - is an activity activated when a process requires new, not yet allocated, memory,
 - is a routine called when entering a syscall
- 4) **Main differences between fork() and exec() (choose most complete and significant)**
 - exec() empties the father substituting it with a new process
 - after fork() the two processes are identical
- 5) **Select WRONG sentences about DDS.**
 - In DDS nodes communicate using both the User Datagram Protocol and the Internet Protocol (UDP/IP) unicast and TCP/IP multicast
 - The main goal of DDS is to share the right data at the right place at the right time, even between time-decoupled publishers and consumers
 - DDS applications are interoperable across operating systems, hardware architectures and programming languages
- 6) **Differences between UDP and TCP sockets: select WRONG sentences (multiple choices)**
 - TCP is bidirectional, UDP is one-way,
 - TCP can be used in a same machine, UDP only in the network
- 7) **You have to design a blackboard in a Posix machine and you have to privilege speed. Choose one answer**
 - shared memory with semaphores
 - a TCP centralized socket server
- 8) **To send a signal to a process you must**
 - use kill() but only to a process that you own
- 9) **A publish/subscribe architecture**
 - Promotes the usage of a modular design
- 10) **Two sensor systems in a robot generate asynchronously single data that a monitor has to process. Instead of a select() method, we use this scheme:
As soon as a sensor writes its datum into a pipe, it sends a signal to inform the monitor. When the monitor reads the datum, it sends in its turn a signal to the sensor. A sensor can send again its datum if necessary. Important fact: sometimes both the monitor and the sensors may mask signals. Choose the best sentence that describes the correctness or the drawback(s) of this solution with respect to the select() method.**
 - non correct, data may be lost and pipes may become full with no reason
- 11) **Can two processes share memory?**
 - Yes by using mmap system call
- 12) **Here is a list of objects that can be referenced by a file descriptor or generate a file descriptor. Select the two items who DO NOT**
 - u-area,
 - PID,

■ **signal number**

13) when a process accesses a file on disk several data structures of the operating system are involved:

1. file descriptors in the U-Area
2. file table in the kernel memory
3. i-node table in the kernel memory
4. disk cache
5. users table
6. socket status

But one of these is wrong! Which one?

■ **6**

14) Windows' and OSX' RPC is a system call that can be implemented in Posix using

■ **TCP client / server model**

15) Choose the best definition for Posix non determinism

■ **a concurrent program whose execution does not have a predetermined sequence, depending on the interaction with other processes**

16) Posix signals can stop/resume a process

■ **only using special signals**

17) main characteristics of the the fork() exec() model are (choose 3 answers max)

- **create two concurrent independent processes**
- **can create two processes not absolutely identical**
- **do not duplicate all data oppure l'altro create**

18) Choose 4 RIGHT sentences about named and unnamed pipes.

- **both pipes are fifo,**
- **named pipes are on disk,**
- **named pipe: multiple process can connect dynamically / disconnect,**
- **named pipes can be used for a service different than fifo IPC**

19) What IPC paradigms are missing in Posix (or non directly available)? (you can choose more than one answer)

- **publish / subscribe messaging,**
- **remote procedure calls**
- **blackboards**
- **events with sender's name**

20) If a process is suspended, waiting on a pipe, a Posix signal can awake it (choose the best)

- **True**
- **Of course in strict sense is false; one piece of the process can be awakened**

21) A process has multiple pipes to choose reading from; which is the best thing to do? It can (choose the best answer)

■ **choose randomly a pipe with data, or wait a limited time**

22) select() is used to (choose the best answer)

■ **choose which pipes can be read immediately out of a set of pipes or wait a given time**

23) Posix signals can change the process (scheduling) status? Select a WRONG answer.....

- **yes, but only lowering the process priority,**
- **never questa**

24) using file flock() you can implement a semaphore?

■ **yes, binary or only between a processes**

25) A process can send a signal only to

■ **itself, its descendants, all processes of the same owner**

- 26) The Posix pselect() does NOT have as parameter: (choose multiple answers with the parameters NOT used by select)
- process ID(s),
 - pointer to signal handler,
 - seed number for random choice of fds to read or write
- 27) The swapper daemon has the aim of (choose the most complete and significant)
- extending the main memory to mass storage to implement the virtual memory model
- 28) the exec() system call (choose the most complete)
- erases all data and code segments, but preserves open file descriptors
 - runs a new process with a new PID
- 29) The file definition in Posix has a general meaning for defining several communication models and primitives. One of the following is wrong: which one?
- signal raising and signal handling
- 30) How can be defined the shell? Choose the wrong sentences
- a process that defines mandatory scheduling parameters of the executables you run,
 - a process for editing, compiling, running programs
- 31) choose the best and correct sentence about the shell
- an interactive user process interpreting and executing commands
- 32) A client / server socket has the following states (setsockopt is optional):
The connect state at client side is reached after the server is in state "socket" (a), "bind" (b), "listen" (c) or "accept"(d) ?
- D
- 33) What is the exit status of a process?
- an integer number captured by the parent upon some child's termination by wait() or waitpid()
- 34) in a machine there are 3 processes: A1 and A2 perform data acquisition, and P collects. Data are simple, very rare but urgent (must be caught as soon as possible). Which primitives do you choose to do that in the most efficient way?
- pipes and signals
- 35) What's the right difference between open() and creat() when you want to read from a file?
- if the file does not exist, no difference (but you need to add parameters to open)
- 36) After a fork() two processes share (choose the best and complete answer)
- current directory, signal action, accounting information, open file descriptors, and some PIDs info
- 37) Signals can be used as Inter Process Communication (IPC) primitives for exchanging data between processes
- no: theoretically they might, but difficult to manage in POSIX
- 38) You should use the errno variable for
- knowing the error occurred after a system call but only if an error occurred
- 39) The basic (non RT) posix scheduler (select 3 answers MAX)
- is a round robin with priority,
 - alters automatically the priority of a process by lowering it during execution
 - is a function started by any event (e.g. interrupt, syscalls, signals..)
- 40) A signal mask is (choose the most complete)
- the set of signal numbers blocked in a process, stored as bitset
- 41) a file descriptor may denote:
1. a file on disk
 2. a directory in the file system
 3. a device
 4. a shared memory
 5. a select

- 6. an IPC stream channel
 - 7. a socket in network
 - 8. a pseudofile, e.g. for accessing kernel values
 - 9. a pseudostream for reading and writing (e.g. random numbers, null output)
 - 10. a semaphore
- But two are wrong! which ones?

■ 5

■ 10

42) A file descriptor in posix can NOT be used for: (choose the answer with the WRONG concept)

■ random number generators

43) A file descriptor in posix can be used for: (select a WRONG answer)

■ identify processes PIDs

44) a file descriptor in Posix

■ denotes a raw stream of data and is defined by an open() and several related primitive(s)

■ connects processes to regular file or to other processes for serial (streaming) communication (forse)

45) A daemon, or service is

■ a periodic process that manages memory, network, time, disk and similar system wide services

46) Choose WRONG sentences about the U-Area (multiple answers)

■ is a data set with all information about the process upon its execution and never changes,

■ is a data set about the process which is duplicated after a fork but not after an exec,

■ is a data set about the process which is duplicated after an exec but not after a fork,

■ is a data set with the process status used by the scheduler

47) You have to design a blackboard in the net using Posix and you have to privilege speed. Choose max two answers

■ a TCP socket server ,

■ an UDP socket server

48) Signals can come (choose the most complete)

■ from any other process of the same owner, from the kernel, from the process itself

49) Using a POSIX shared memory with two processes, why semaphores are needed? Choose the most specific answer.

■ when mutual exclusion is necessary regarding access of memory locations

50) A process image is composed of Data, Stack, Executable Code and User Area; one of the 4 parts may be equal or shared with respect to other processes activated by a fork(). Which one?

■ data are equal after fork() and may be shared using shm_open()

51) The User Area of a process does not contain (choose all correct sentences)

■ parent ID

■ process ID

■ process priority

52) In this scheme of a TCP socket session: what you put in the four missing blocks, in the proper order (upper left, upper right, lower left, lower right)?

■ accept connect close close

53) main difference between signal and h/w interrupt (choose two answers)

■ signals are raised by software statements, interrupts by the electronics, but have similar effect

■ a user process can mask signals, not interrupts

■ signals may have a code to be executed in user space, interrupts have in kernel space

54) The simplest and correct definition of a socket (which defines its behaviour, its aim and its use) is

■ a bidirectional fifo between two dynamically connected processes in the net

- a serial communication model that merges computer and networking principles

55) Choose the right sentences

- UDP allows simultaneous multiple clients
- UDP is faster

56) In a non RT kernel, a process' priority varies periodically; which is the basic default formula for user processes priority?

- A) new priority = base value - average CPU usage in the last second
- B) new priority = base value + integral CPU usage since the beginning of the execution
- C) new priority = base value + $1/2 \times$ CPU usage in the last second
- D) new priority = previous value + $1/2 \times$ CPU usage in the last second

■ C

57) select a wrong sentence

- neither interrupts, system calls and signals activate the scheduler (switcher)

58) after a fork() the process data are

- copied but not shared in the child process

59) if you cannot use lseek(), can you accomplish indexed reading using other system call(s)?

- repeated close(), open(), read() discarding read values

60) In a concurrent architecture one task (producer) sends the same stream of data to 3 consumers, and the consumers read all data very quickly, with minimum delay. Data are packets of 100 bytes. Which is the most suitable (efficient, simple) Posix primitive?

- Shared memory

61) a good definition of non determinism in concurrent programs (distributed, composed of many exec()) is

- non determinism guarantees that the results are independent by the computing speed of every process

62) Steps for establishing a TCP socket from the client side are

- Socket() system call, connect() system call to the address of the server, send and receive data using read() and write() system calls

63) A publish/subscribe architecture

- Promotes the usage of a modular design

64) For managing the memory two demons are used: swapper and pagedemon. What is the main difference?

- swapper rearranges memory to efficiently locate space for the processes; pagedemon frees the memory when processes need space

65) After execution of sleep-family system call, the calling process is put in

- asleep state

66) a signal handler (choose right answers)

- is a function executed once upon receiving a signal
- is a function whose execution interrupts the execution of a process which then it goes back to the interruption point
- is a function pointed by signal() or sigaction() syscalls

67) a signal handler.... (choose the most complete and correct answer)

- is a C function executed upon receiving a specific signal by the process

68) how can link() be properly defined?

- creates a synonym for a file

69) can unlink() delete a file?

- yes, if the file after creation was never further linked using link()

- 70) select right sentences about a system call
- logically acts like an interrupt service routine
 - on return, gives control to the switcher
 - any program executes system calls, at least one
- 71) Shared memory in Linux/Unix can be used for
- implementing IPC through files: mapping files into memory
- 72) after an exec() the child data are
- file descriptors are copied
- 73) POSIX is designed to assure portability of a certain code, not only in robotic application. This is based on the fact that
- in every POSIX compliant machine you can compile and execute the same code
- 74) The perror() system call should be used (choose best answer)
- after checking the return value of a system call
- 75) a context switch (by the scheduler, or switcher) does occur surely when
- syscall is requested
- 76) after process termination
- the father can read a return value decided by son
- 77) a father process can widely communicate/interact with its son (choose the best sentence)
- during execution, using file descriptors
- 78) After a select() syscall, a process changes state:
- from running to asleep
- 79) Do fflush() and fsync() do the same?
- fflush() acts on a single stream, fsync on all streams
- 80) After any open() a new file descriptor is generated... which one?
- the lowest integer number available (not used) in the file descriptors list in user area
- 81) The process table (choose a wrong sentence):
- is kept in the User Area of a process
- 82) The process table stores many data. At least one of the following is wrong. Which ones? (select multiple answers)
- IP address
 - received signals
 - Open file descriptors
- 83) Communication through named and unnamed pipes have different time efficiency? Choose the best answer.
- no, both use main memory as fifo buffers
- 84) POSIX is a standard for program portability in
- Linux/Unix, MAC OS, Windows and others (requires recompilation)
- 85) choose the right sentences regarding signals
- sigkill and sigstop cannot be ignored
 - some signals change the process status (perform context switch)
- 86) fork() and exec() are similar because (choose the most significant wrong answers)
- both erase all data and code segments
 - both generate a new process with a new PID
 - they allow passing numeric parameters to the newly executed process
- 87) In a real robot software, is memory swapping useful? Select the more complete answer.
- No: lower speed, lower predictability of execution time, hardware stress
- 88) Differences between UDP and TCP sockets: select RIGHT sentences (multiple choices)
- in UDP there are no accept(), nor connect()
 - UDP does not guarantee the errorless exchange of data

- For a reliable data transfer (e.g. for remote operation of a drone) you must include an error checking algorithm in the server / client processes

89) Pipe, alone, is a primitive for easily implement IPC through

- blocking fifo raw data

90) The swapper daemon is always running? It does what?

- Can stay inactive if swapping is disabled (frequent in robotics) questa

91) A client / server socket has the following states (setsockopt is optional):

The connection at client side is reached after the server has accepted. When the server closes the connection (not shown in figure) the client reads / receives

- END-OF_FILE

92) Choose the correct statements. While typing the names of folders and files in the shell ...

- pressing TAB key autocompletes the name of the folder/file.

93) The cyclic processes A and B share the memory and use semaphores. A executes two parts, say "some code 1" and "some code 2". The two parts require many seconds. Process B, every T seconds (less than the execution time of the two parts), checks if the process A is executing either "some code 1" or "some code 2". This is the chatGPT's solution, but it is INCORRECT.

Semaphore mutex = 1

Semaphore code1Executing = 0

Semaphore code2Executing = 0

-- Process A

while true do {

wait(mutex)

signal(code1Executing)

signal(mutex)

-- Execute "some code 1" -- This may take many seconds

wait(mutex)

signal(code1Executing)

signal(mutex)

wait(mutex)

signal(code2Executing)

signal(mutex)

-- Execute "some code 2" -- This may take many seconds

wait(mutex)

signal(code2Executing)

signal(mutex)

}

-- Process B

while true do {

sleep(T)

wait(mutex)

if value(code1Executing) == 1 or value(code2Executing) == 1 {

-- Process A is executing either "some code 1" or "some code 2"

-- Perform actions or take necessary steps }

signal(mutex)

}

Select THE RIGHT SENTENCES

- wait(code1Executing) and wait(code2Executing) are missing: introduce to solve

- **semaphores code1Executing and code2Executing can be substituted by boolean variables set to TRUE and reset to FALSE: introduce to solve**

94) Two cyclic producer processes generate random numbers and send them to a consumer process, under some conditions.

The consumer process takes into account an incoming number under additional conditions.

Processes share memory.

Select the best answer. Do we need semaphores? where?

```
procedure producer1() {
    while (true) {
        randomNumber = generateRandomNumber()
        if (randomNumber > lastNumberByProducer2) {
            sendNumberToConsumer(randomNumber)
            lastNumberByProducer1 = randomNumber
        }
        sleep(someTime)
    }
}

procedure producer2() {
    while (true) {
        randomNumber = generateRandomNumber()
        if (randomNumber > lastNumberByProducer1) {
            sendNumberToConsumer(randomNumber)
            lastNumberByProducer2 = randomNumber
        }
        sleep(someTime)
    }
}

procedure consumer() {
    while (true) {
        receivedNumber = receiveNumberFromProducer()
        if (receivedNumber > lastReceivedNumber) {
            processNumber(receivedNumber)
            lastReceivedNumber = receivedNumber
        }
    }
}
```

- **couples of mutex are necessary before row 4/7, 15/18**

95) A process P produces data and consumer C reads them through shared memory. They are concurrent and use a double buffer: two arrays (say 0 and 1). P writes in one of them, while C reads from the other. When a buffer has been written and the other one read, the two buffers are swapped.

The algorithm looks like:

semaphores S1 = A S2 = B M = C

P::

buffer number = 1

while(1) {

wait S2

wait M

swap buffer number


```

signal M
write in buffer until full
signal S1
}
P::
buffer number = 0
while(1) {
wait S1
wait M
swap buffer number
signal M
read from buffer until empty
signal S2
}

```

Choose the right values for A, B, C

- A 0 B 1 M 1
- A 0, B 1, M useless (avoid its use) questa

96) A special producer / consumer allows the producer to overwrite the produced value before the consumer reads it. In this case the consumer loses one or more produced values, but cannot read two times the same value until a new one is produced. Consider the following pseudocode:

shared variables:

```

int n;
semaphore s = 0;
Producer:
int pint;
while (1) [
    produce pint;
    n = pint;
    signal(s);
]

```

Consumer:

```

int cint;
while (1) [
    wait(s);
    cint = n;
    use cint;
]

```

Is it correct or not?

- Almost right, another semaphore should be used for mutual exclusion

97) A process reads data from two file descriptors, say fd1 and fd2, data from two producers. This is the pseudocode:

```

while (TRUE) {
select fd1 or fd2 or both
switch (data available?) {
    execute in random order {
        case (fd1 has data): fd = fd1, read (fd, data), break
        case (fd2 has data): fd = fd2, read (fd, data), break
    }
}
}

```

```

    }
    default: suspend execution for timeout seconds, waiting for data
}

```

We want to implement in POSIX the pseudocode fragment above, using `select()` with very unlimited timeout value. What is the consequence?

- **OK, but you cannot manage the possibility that the two producers stopped sending data**

98) A process P produces data and consumer C reads them through shared memory. They are concurrent and use a double buffer: two arrays (say 0 and 1). P writes in one of them, while C reads from the other. When a buffer has been written and the other one read, the two buffers are swapped. The algorithm looks like:

semaphores S1 = 0 S2 = 1

P::

buffer number = 1

```

while(1) {
    XXX
    swap buffer number
    write in buffer
    YYY
}

```

}

P::

buffer number = 0

```

while(1) {
    TTT
    swap buffer number
    read from buffer
    ZZZ
}

```

}

Choose the right values for XXX, YYY, TTT, ZZZ.

- **XXX wait S2, YYY signal S1, TTT wait S1, ZZZ signal S2**

99) This is a pseudo code that describes reading data from two processes through pipes 1 and 2. Reading must be non deterministic, hence without blocking, and must assure fairness. What should be written in XXX?

fd1 = open (pipe 1), fd2 = open (pipe2);

```

while (TRUE) {
    select fd1 or fd2 or both
    switch (data available?) {
        case (fd1 has data && fd2 has data):
            choose [how ?] fd, read (fd, data), break
        case (fd1 has data):
            fd = fd1, read (fd, data), break
        case (fd2 has data):
            fd = fd2, read (fd, data), break
        default: suspend execution for a given time
    }
}

```

- **randomly using a random number generator of uniform distribution**

100) This is a pseudocode fragment:

```
semaphore k=1
process 1::
while(1) [
  print "1a"
  wait(k)
  print "1b"
  signal(k)
]
process 2::
while(1) [
  print "2a"
  wait(k)
  print "2b"
  signal(k)
]
```

To force a sequence like "1a2a1b2b1a2a1b2b..." are changes necessary? (Choose the best answer)

■ **2 semaphores, k1 and k2, with different initialisations, cross-activated by the two processes**

101) Pseudocode using shared memory and wait/signal semaphores.

In this text a scheduler activates two cyclic processes. Suppose that "Code For ProcessA" and "Code For ProcessB" have access to shared variables.

The question is: are we sure that process A and process B have exclusive access to those variables? Choose the most complete and exhaustive answer.

```
semaphore processA = 0
semaphore processB = 0
integer BA = BB = 0
procedure ProcessA() {
  while (true) {
    wait(processA)
    BA = 1
    // Code for ProcessA's execution
    BA = 0
    wait(processA)
  }
}
procedure ProcessB() {
  while (true) {
    wait(processB)
    BB = 1
    // Code for ProcessB's execution
    BB = 0
    wait(processB)
  }
}
procedure Scheduler() {
  while (true) {
    // Schedule ProcessA
    if(BB == 0) signal(processA)
```

```

sleep(someRandomTime)
// Schedule ProcessB after someTime
if(BA == 0) signal(processB)
// Allow some time
sleep(someRandomTime)
}
}

```

■ **Not likely**

102) A process MASTER controls executions of processes A and B using semaphores. Process A iterates 3 times an inner code. When process B starts it temporarily stops A and resumes it upon ending. The user is requested for commands: startA, startB. This is the pseudocode:

```

semaphores
S1= 0, S2=0, S3=3
A::
while(1) [
wait s1
i=3
while (i-->0) [
wait S3
<execute A's action> ]
S3=3
signal s1
]
B::
while(1) [
wait s2
wait S3
<execute B's action>
signal S3
]
MASTER::
while(1) [
input command
switch command
case startA: signal s1; break
case startB: signal s2; break
]

```

The algorithm is wrong or good?

■ **wrong: A is not interrupted by B except some cases**

103) Pseudocode for shared memory and wait/signal semaphores. Here a producer/consumer algorithm is depicted, which has errors. You are

asked to tell which are the wrong lines; you shall select all wrong lines (multiple choices).

1 shared buffer of size N

2 semaphore mutex = N

3 semaphore fullSlots = 0

```
4 semaphore emptySlots = 1
5 procedure producer() {
6 while (true) {
7 produceItem(item)
8 wait(fullSlots)
9 wait(mutex)
10 insertItemIntoBuffer(item)
11 signal(mutex)
12 signal(emptySlots)
13 }
14 }
15 procedure consumer() {
16 while (true) {
17 wait(fullSlots)
18 wait(mutex)
19 item = removeItemFromBuffer()
20 signal(mutex)
21 signal(emptySlots)
22 consumeItem(item)
23 }
24 }
```

■ 4

■ 2

■ 8 and 12